



FRANCESCO BALDUCCI

Artificial Intelligence Engineer and Machine Learning Enthusiast

I am a curious and results-driven engineer specializing in Cognitive Science through my Master in Human-Centered AI at DTU. With a background in full-stack development and cognitive science, I am eager to work on challenging end-to-end machine learning pipelines and their statistical evaluation. I am especially passionate about the connection between advanced ML and human-centered AI experiment design.

RECENT PROJECTS

Personalized Multi-Modal Agent System for Bias Analysis

Engineered an end-to-end agent for YouTube bias analysis through RAG, LangChain and LangGraph. Curated the embedding processing of a hybrid text/image dataset in a Chroma vector database. Evaluated the performance of each component with a custom LLM-as-Judge in an ablation study.

Node Centrality and Link Prediction in a Social Graph

Analyzed centrality in a social network with Brandes' Algorithm for Betweenness and PageRank. Also benchmarked GCNs and Spectral Clustering for link prediction by splitting datasets to evaluate accuracy and explainability. Tuned hyperparameters (learning rate, k-eigenvectors) to select best method, achieving 89.3% accuracy.

EEG Signal Processing for Motor Imagery Classification

Built a pipeline for EEG motor imagery classification. Filtered relevant bands via Notch/Band-pass filters, extracting features using supervised CSP and STFT band-averaged magnitudes. Compared performance of FNN and a CNN in a k-fold cross-validation setup obtaining up to 75% accuracy.

Manipulation of Belief Effects on AI-Generated Content

Executed an experiment on emotional responses to GenAI labels of videos. Collected HRV and SAM data from 22 participants including ethical considerations, consent collection and stimuli randomization to suppress confounding factors. Finally conducted Shapiro-Wilk and Wilcoxon signed-rank tests (via bootstrapping).

My Projects:

[balduccifrancesco.github.io](https://github.com/balduccifrancesco)

Contacts:

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AI & ML skills:

Python	SVM
PyTorch	Classification
LLMs	Regression
LangChain	Clustering
LangGraph	k-means
MM-RAG	Decision trees
VectorDBs	AdaBoost
Transformers	k-nearest neigh.
FNN / MLP	Naive Bayes
CNN	Density estimation
RNN	Outlier detection
GCN	Signal Processing
PCA	Graph Analysis

Software & DevOps skills:

Java	SQL / NoSQL
JavaScript	Git
REST APIs	Docker
Spring Boot	Azure DevOps
React / Angular	Agile / Scrum

Statistical & Experimental skills:

Conf. intervals	Cross-validation
p-value	Data visualization
ANOVA	PsychoPy
Wilcoxon test	Pandas
Shapiro-Wilk test	NumPy

Language skills:

English	CEFR C1
Italian	native speaker
Danish	beginner

PROFESSIONAL EXPERIENCES

Full-stack web development • Whitehall AI Reply, Rome, Italy

January 2024 – December 2024

Consulting activities as developer on multiple projects for central government agencies. *Main activities:*

- JAX-RS / SpringBoot back-end development involving RESTful APIs and SOAP, following the MicroProfile model. Manipulation of Oracle DBMS and utilization through JPA / Hibernate;
- React front-end development connected to back-end APIs;
- Daily interaction with the client for requirements gathering and release of new features following the AGILE methodology;

Angular front-end web development • Telebit SPA, Treviso, Italy

March 2021 – July 2021, March 2022 – December 2023

Creation of a management web platform through agile development cycle (SCRUM). *Main activities:*

- Task scheduling and CI/CD through Azure DevOps services such as Git repositories, artifacts, backlogs and tasks;
- Use and customization of libraries such as PrimeNG and Angular Material;
- Implementation of a Design System starting from Figma mockups;

EDUCATION AND TRAINING

MSc Eng Degree: Human-Centered AI

February 2025 – Current

Human-Centered Artificial Intelligence @ DTU (Denmark). *Relevant Coursework:*

- 02806 Social data analysis and visualization
- 02411 Statistical design and analysis of experiments
- 02456 Deep learning
- 02464 Artificial intelligence and human cognition
- 02455 Experiment in cognitive science
- 22053 Principles of brain computer interface

Overseas Exchange Student

February 2026 – July 2026

Exchange Semester @ University of Queensland (Australia). *Relevant Coursework:*

- INFS4205 Advanced Techniques for High Dimensional Data
- INFS7450 Social Media Analytics

«ReSP» course (soft skills and humanities)

October 2020 – July 2023

Cohabitation and personal growth activities @ ELIS residence, Rome, Italy